ATNA

Air Traffic Network Analysis

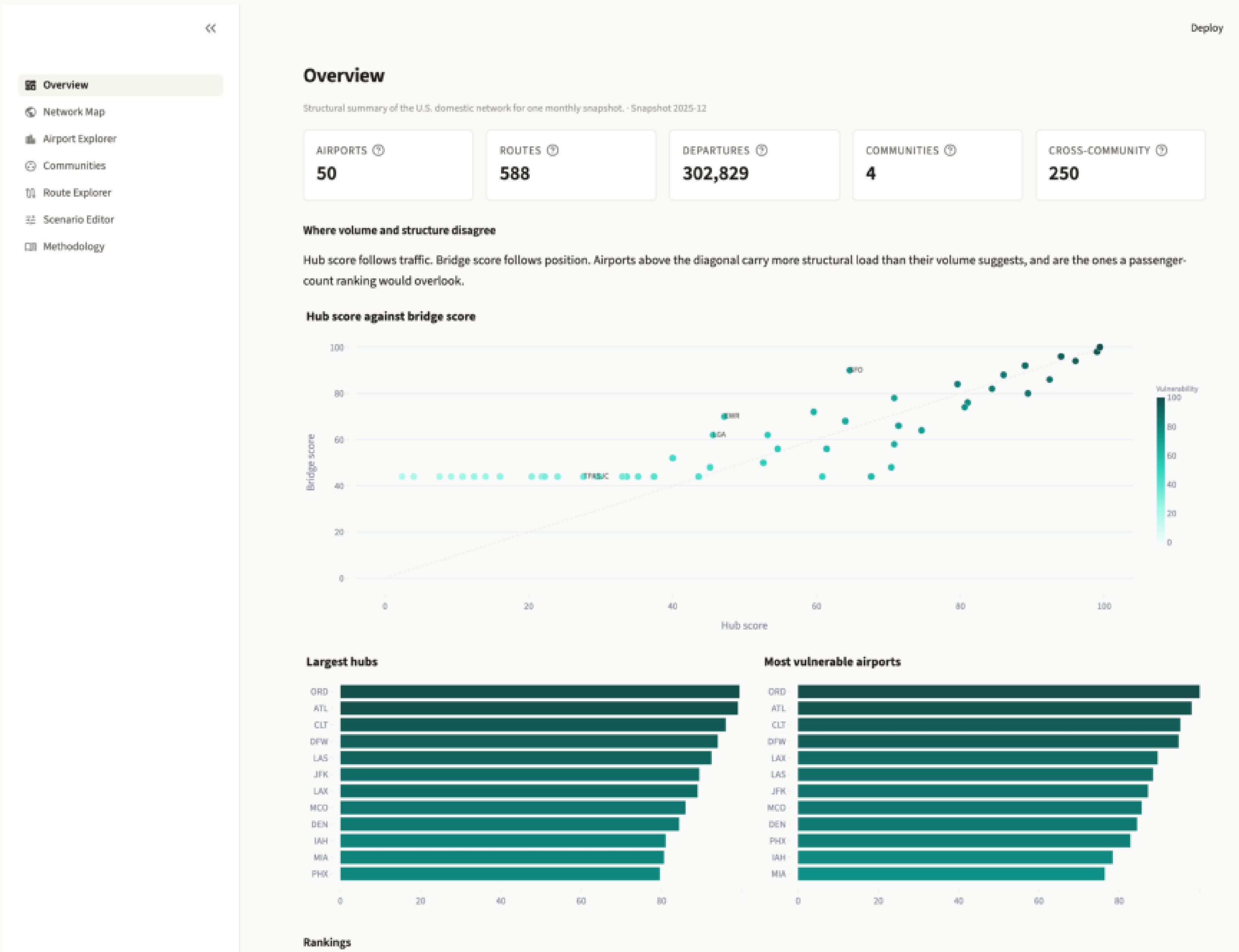
Interface walkthrough

Structural criticality and scenario modelling for the U.S. domestic flight network. Every panel is a live capture of the running application.

Snapshot	2025-12
Pages	8
Generated	2026-08-24

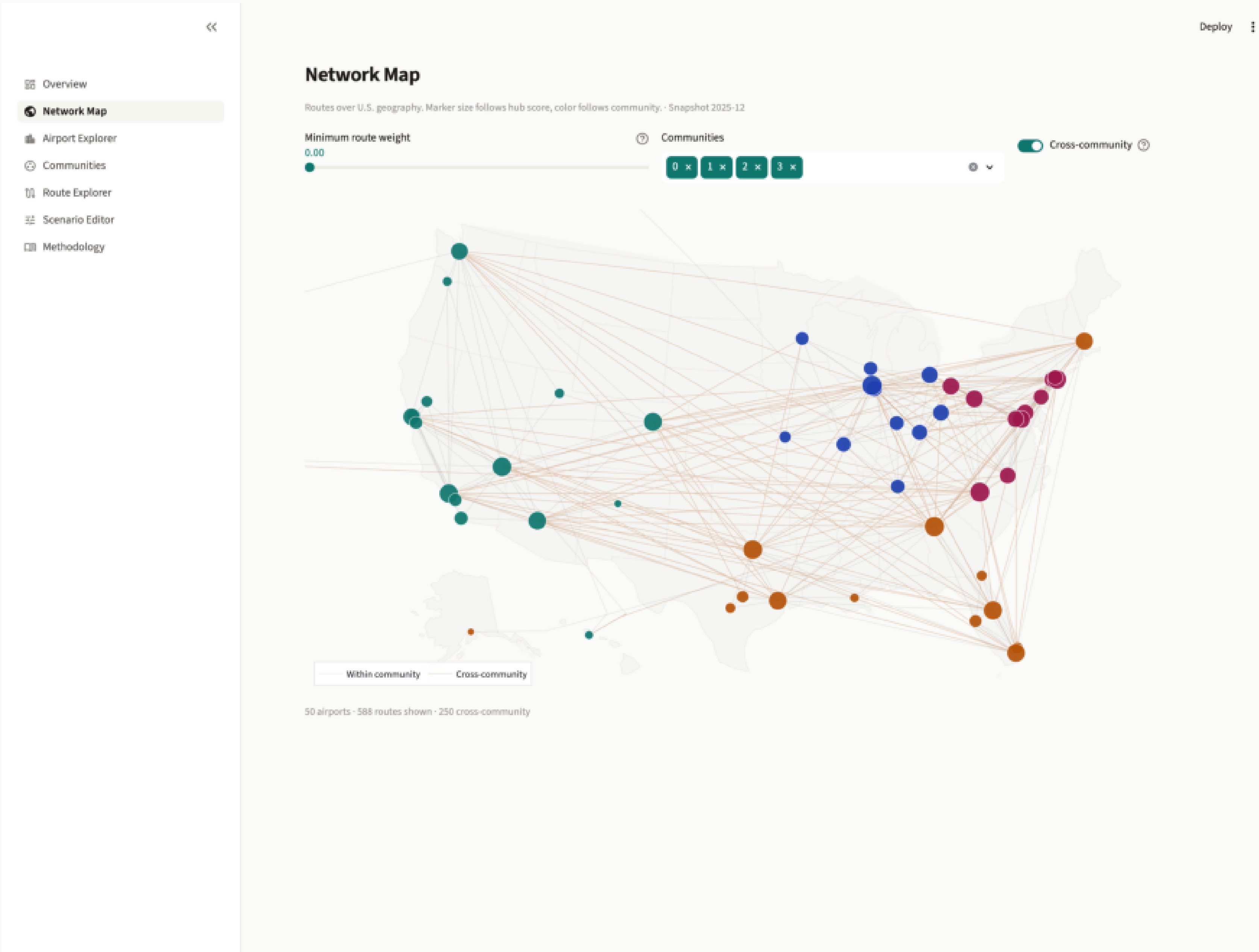
Overview

Snapshot totals, then the projects central claim: hub score against bridge score with the diagonal drawn in. Airports above the line carry more structural load than their traffic implies.



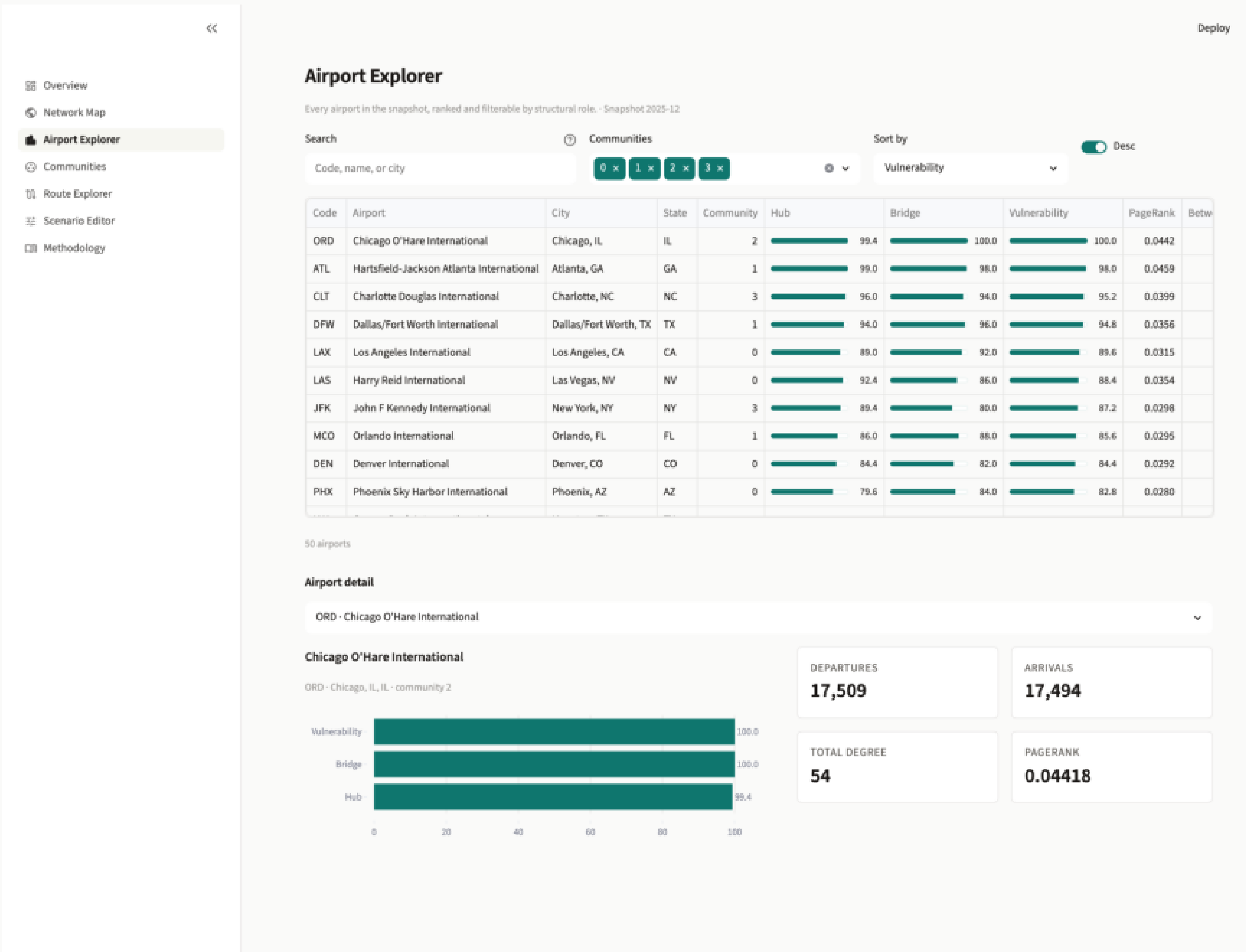
Network Map

The route network over an Albers USA projection. Marker area follows hub score, color follows Leiden community, and cross-community routes are tinted separately. The regional structure is legible directly from the map.



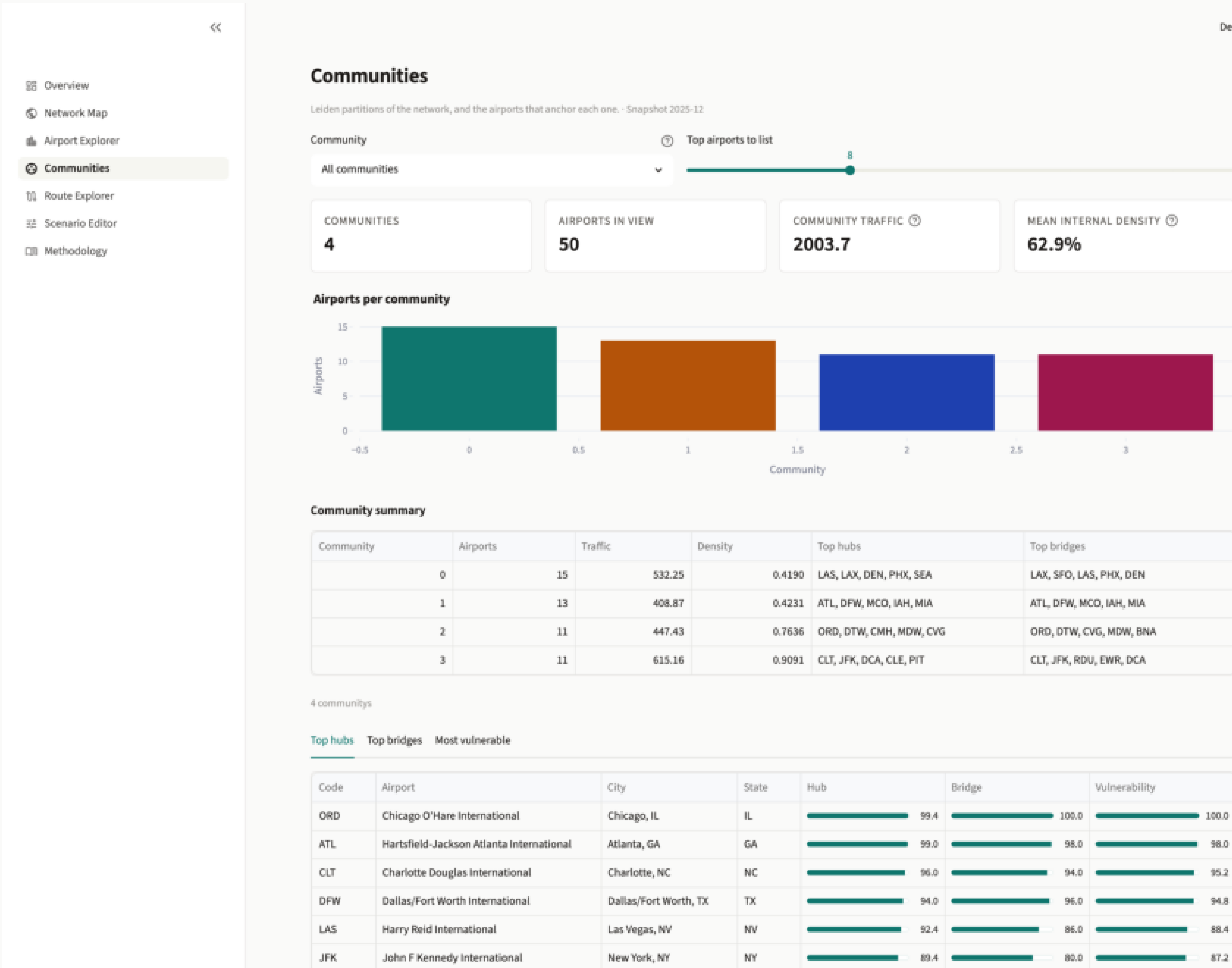
Airport Explorer

Every airport in the snapshot, searchable by code, name, or city. Percentile scores render as bars on a fixed 0-100 domain so standing is comparable between tables.



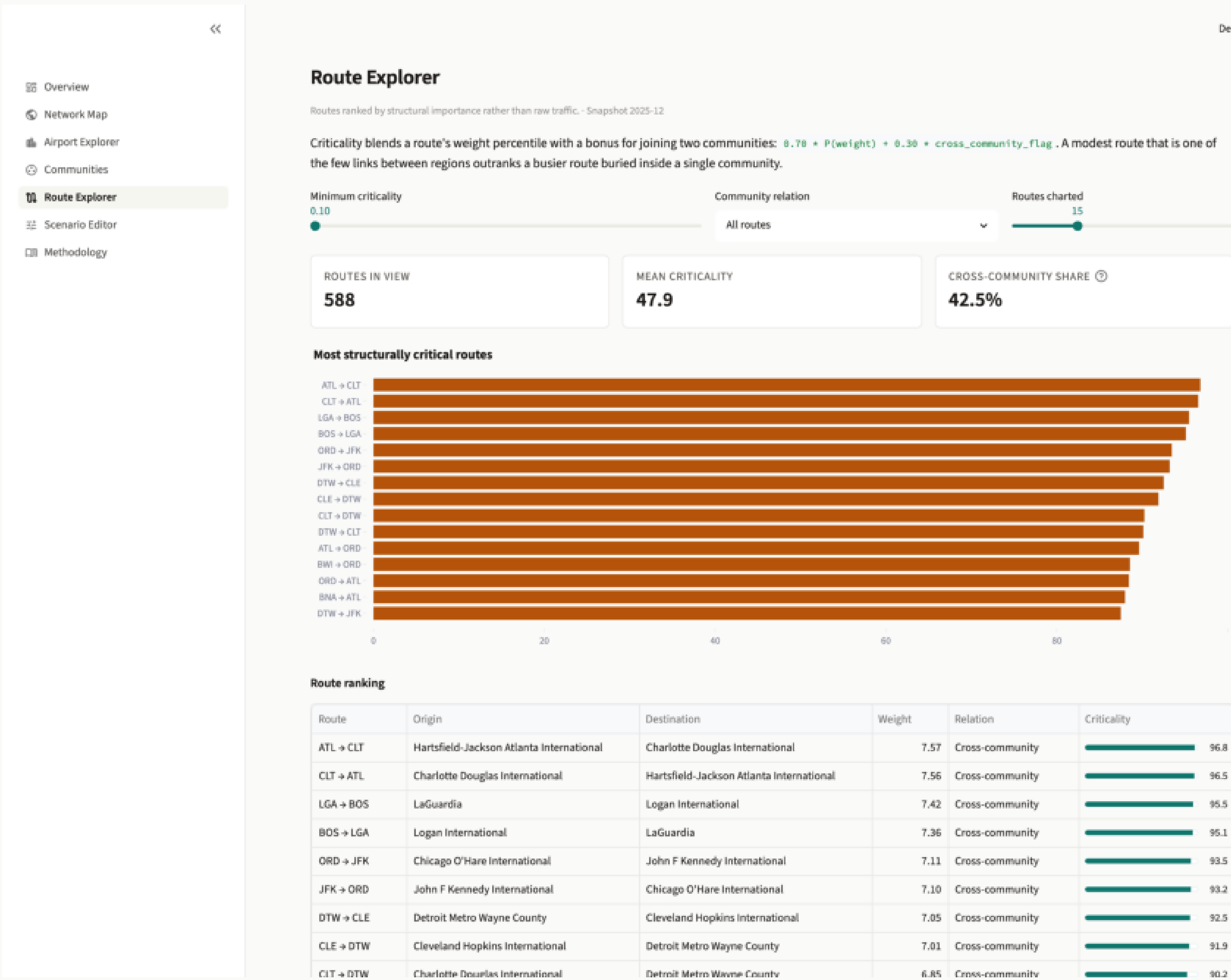
Communities

Leiden partitions with their size, internal traffic, and density. The top hub and bridge lists stored as airport ids in the artifact are resolved to codes for display.



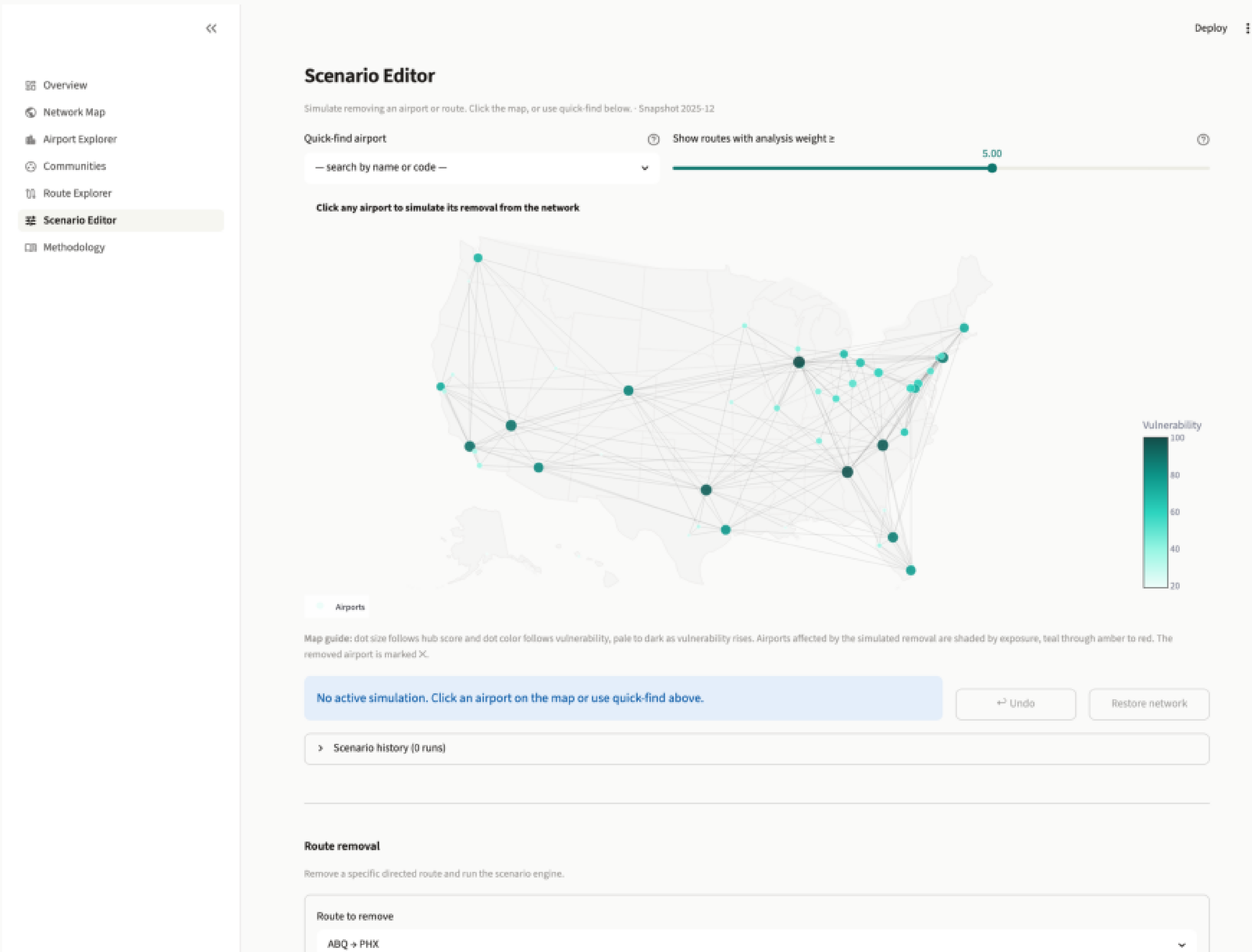
Route Explorer

Routes ranked by structural importance rather than raw traffic. A modest route that is one of few links between regions outranks a busier route buried inside one community.



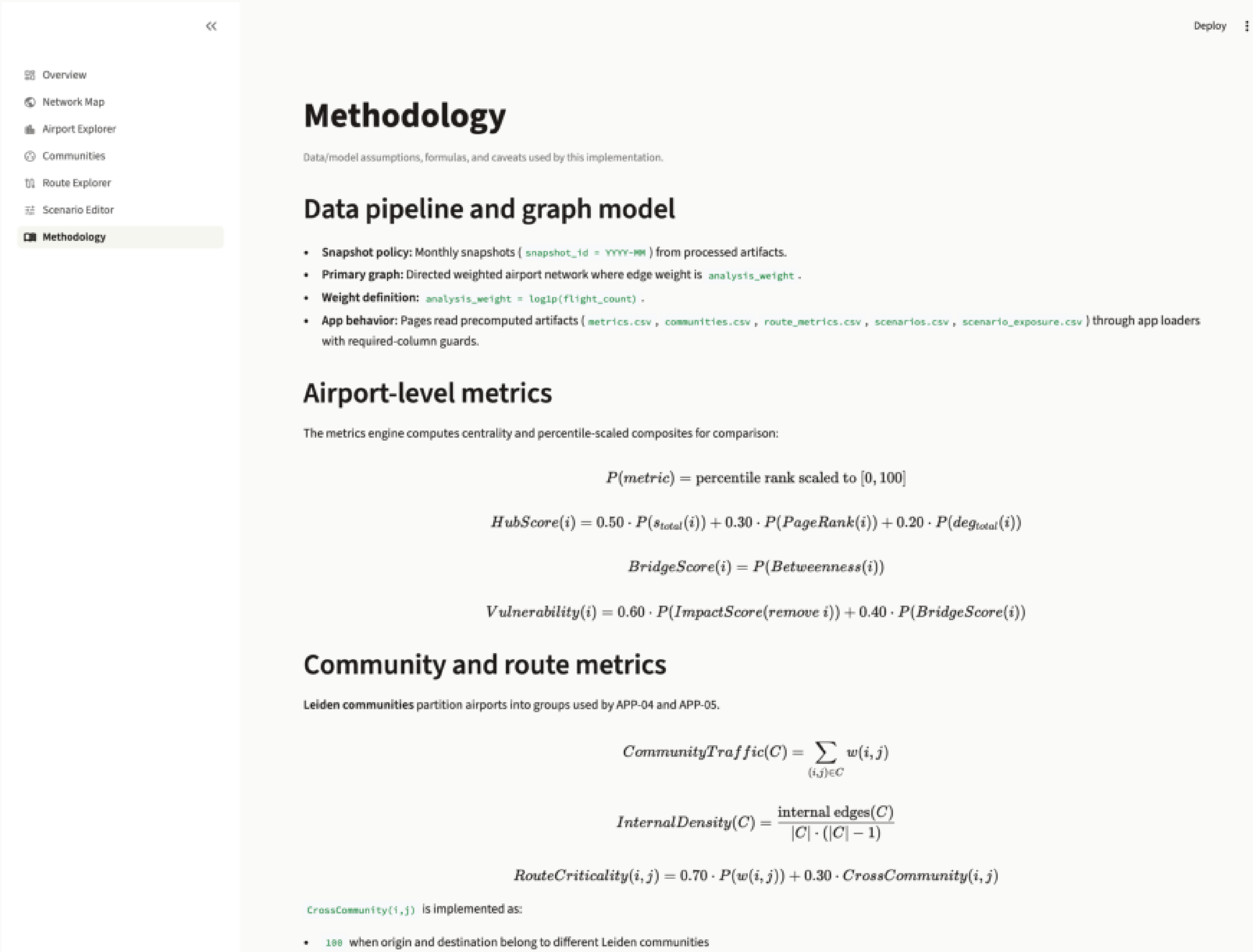
Scenario Editor

The baseline network before any simulation. Any airport can be removed by clicking it, or through the quick-find control.



Methodology

Every formula as implemented, followed by the model's known limitations stated plainly rather than omitted.



Deploy

Overview
Network Map
Airport Explorer
Communities
Route Explorer
Scenario Editor
Methodology

Methodology

Data/model assumptions, formulas, and caveats used by this implementation.

Data pipeline and graph model

- **Snapshot policy:** Monthly snapshots (`snapshot_id = YYYY-MM`) from processed artifacts.
- **Primary graph:** Directed weighted airport network where edge weight is `analysis_weight` .
- **Weight definition:** `analysis_weight = log1p(flight_count)` .
- **App behavior:** Pages read precomputed artifacts (`metrics.csv` , `communities.csv` , `route_metrics.csv` , `scenarios.csv` , `scenario_exposure.csv`) through app loaders with required-column guards.

Airport-level metrics

The metrics engine computes centrality and percentile-scaled composites for comparison:

$$P(metric) = \text{percentile rank scaled to } [0, 100]$$

$$HubScore(i) = 0.50 \cdot P(s_{total}(i)) + 0.30 \cdot P(PageRank(i)) + 0.20 \cdot P(deg_{total}(i))$$

$$BridgeScore(i) = P(Betweenness(i))$$

$$Vulnerability(i) = 0.60 \cdot P(ImpactScore(remove\ i)) + 0.40 \cdot P(BridgeScore(i))$$

Community and route metrics

Leiden communities partition airports into groups used by APP-04 and APP-05.

$$CommunityTraffic(C) = \sum_{(i,j) \in C} w(i,j)$$

$$InternalDensity(C) = \frac{\text{internal edges}(C)}{|C| \cdot (|C| - 1)}$$

$$RouteCriticality(i,j) = 0.70 \cdot P(w(i,j)) + 0.30 \cdot CrossCommunity(i,j)$$

`CrossCommunity(i,j)` is implemented as:

- `100` when origin and destination belong to different Leiden communities

Scenario Editor — active simulation

ATL removed. The removed airport is marked, airports touched by the two-hop ripple are shaded by exposure, and the impact and network-health scores update against the live graph.